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Lesson 4: Rational Expressions

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Dividing Rational Expressions

You know that division is the opposite of multiplication. When you divide fractions, you invert the fraction you are dividing by and then multiply.

Example 1

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Divide: $\frac{7}{12} \div \frac{5}{6}$

Step 1

Change to multiplication and invert the divisor.

$$\frac{7}{12} \div \frac{5}{6} = \frac{7}{12} \times \frac{6}{5}$$

Step 2

Cancel and multiply.

$$\frac{7}{\cancel{12}_2} \times \frac{\cancel{6}^3}{5} = \frac{7}{10}$$

Dividing rational expressions works the same way. Change to multiplication and invert the second expression.

Example 2

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Divide: $\frac{m^3n}{20n} \div \frac{2mn}{15mn^3}$

Step 1

Change to multiplication and invert the divisor.

$$\frac{m^3n}{20n} \div \frac{2mn}{15mn^3} = \frac{m^3\cancel{n}}{\cancel{20n}_4} \cdot \frac{\cancel{15mn}^3n^2}{2\cancel{mn}_1}$$

Step 2

Cancel.

$$\frac{m^3\cancel{n}}{\cancel{20n}_4} \cdot \frac{\cancel{15mn}^{3n^2}}{2\cancel{mn}} = \frac{3m^3n^2}{8}$$

Step 3

Multiply straight across.

$$\frac{m^3\cancel{n}}{\cancel{20n}_4} \cdot \frac{\cancel{15mn}^{3n^2}}{2\cancel{mn}} = \frac{3m^3n^2}{8}$$

Remember to factor each numerator and denominator before canceling. You can cancel identical factors only.

Example 3

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Divide: $\frac{x^2-8x}{x^2+8x+16} \div \frac{x-8}{x^2-2x-24}$

Step 1

Change to multiplication and invert the divisor.

$$\frac{x^2-8x}{x^2+8x+16} \cdot \frac{x^2-2x-24}{x-8}$$

Step 2

Factor.

$$\frac{x(\cancel{x-8})}{(x+4)(\cancel{x+4})} \cdot \frac{(\cancel{x+4})(x-6)}{\cancel{x-8}}$$

Step 3

Cancel.

$$\frac{x(\cancel{x-8})}{(x+4)(\cancel{x+4})} \cdot \frac{(\cancel{x+4})(x-6)}{\cancel{x-8}}$$

Step 4

Multiply across.

$$\frac{x(\cancel{x-8})}{(x+4)(\cancel{x+4})} \cdot \frac{(\cancel{x+4})(x-6)}{\cancel{x-8}}$$

$$= \frac{x(x-6)}{(x+4)}$$

$$= \frac{x^2-6x}{x+4} \quad \checkmark$$

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